# Project Abstract

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## Introduction

Cardiovascular diseases rank among the most common causes of death throughout the world which makes it essential to identify patients at risk for heart disease through early detection methods and preventive measures. Medical professionals use several clinical indicators which include blood pressure and cholesterol levels and heart rate to assess a patient's risk of developing heart disease. The process of assessing these variables at the same time becomes challenging because of its intricate nature. Machine learning techniques provide an opportunity to assist in identifying patterns in medical data and predicting the likelihood of heart disease with high accuracy.

## Project Objectives

The project aims to construct and test multiple machine learning models which will determine whether patients have heart disease. The study will explore how different features such as age, cholesterol levels, resting blood pressure, maximum heart rate, and other clinical measurements contribute to prediction accuracy. The project will evaluate different models to identify which algorithm delivers the most accurate predictions.

## Methodology

The methodology will begin with data preprocessing to handle missing values and to encode categorical variables and to scale numerical features according to their requirements. Exploratory data analysis will be conducted to identify patterns and relationships between variables. The dataset will then be split into training and testing sets. The project will implement and assess various machine learning models which include Logistic Regression and Decision Trees and Random Forest classifiers. Model performance will be evaluated using metrics such as accuracy and precision. Hyperparameter tuning techniques such cross-validation will be used to optimize model performance.

## Dataset

The dataset used in this project is the UCI Heart Disease Dataset, which contains medical attributes of patients and a target variable indicating the presence of heart disease. This dataset is widely used in machine learning research and provides a suitable foundation for predictive modeling in healthcare.